

Vinh Ly

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Summary

Software engineer with **18 years of experience** building and operating systems at Google, YouTube, and Whatnot. Deep experience with **large-scale distributed systems**, **cloud infrastructure**, and **ML pipelines** processing billions of events daily. Spent the last several years working on content rights management, payment systems, and trust/safety infrastructure. Now looking to put that breadth to work as a **Solutions Engineer** helping customers understand and implement privacy and compliance platforms.

Core Skills

Languages: Python, Go, C++, Java, TypeScript, Ruby

Frameworks: Angular, React, Ruby on Rails, Spring Boot

Data: BigQuery, Spanner, MySQL, PostgreSQL, Bigtable, Kafka, Beam, Dagger

Infra / MLE: Kubernetes, Docker, Terraform, TFX, XGBoost, MLflow, uFabric

Cloud: GCP (Compute Engine, GKE, BigQuery, Spanner), AWS (EC2, Lambda, S3)

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Owns and operates Content ID, the rights management system used by over 5 million content owners to register and claim videos on YouTube. The system processes around 200K claims per day and is the primary mechanism protecting copyright holders from unauthorized use of their content.
- ML pipeline owner for YouTube's live stream monitoring, which detects copyrighted audio embedded in live broadcasts in near-real-time. Processes millions of hours of content per day with **sub-5-minute detection latency**, preventing widespread abuse before it spreads.

Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- One of 2 engineers who built Whatnot's payment infrastructure from the ground up, starting before the platform reached \$1B in annualized GMV. Handled payment processing, escrow, refunds, and chargeback workflows across collectibles categories (trading cards, comics, vinyl, sneakers).
- Worked on the Checkout team to improve purchase completion rates and reduce payment failures during high-traffic drops. Later moved to the Trust and Safety team to build detection systems for marketplace fraud and platform policy violations.

Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Led a team of 7 engineers on the Search Frontend migration, rewriting Google's main search UI from a 10-year-old GWT codebase to Angular and Polymer. Ran interleaving A/B experiments throughout rollout, and page load time dropped by roughly 30% for users in the experiment group. Shipped to 100+ countries and 15+ product surfaces.
- Rebuilt the search ranking evaluation pipeline to cut turnaround from 4 hours to under 20 minutes by moving from MapReduce to Dagger and BigQuery. The new pipeline became the standard evaluation tool for the search quality team.
- Led the security monitoring infrastructure for Search traffic, starting with a migration to a new cryptographic library and ending with a real-time anomaly detection system in C++. Built the initial version of what became Google's security monitoring stack for search.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- Software engineer on the Search Frontend team. Contributed to the UI rewrite and worked on backend infrastructure supporting Google's core search results page.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Worked on a team responsible for the core indexing infrastructure behind Bing (then Live Search). Focused on data pipeline reliability and the systems that kept search index freshness at scale.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Summer internship on the same indexing team. Shipped a small internal tool that automated a manual pipeline validation step.